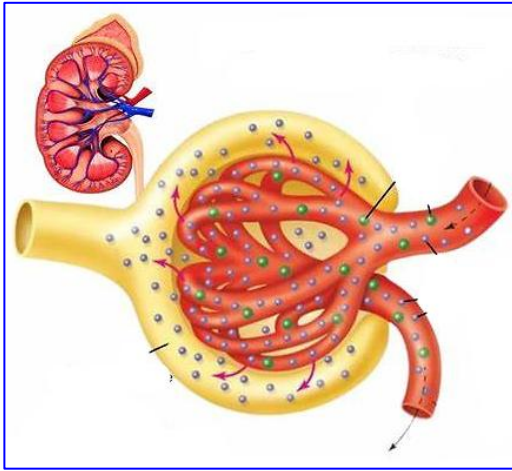
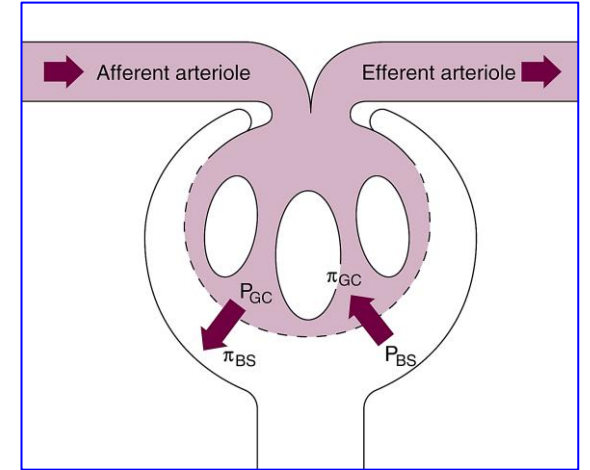




Basic Principles of Medicine 1

Module : Cardio, Pulmonary, Renal

CPR: Lecture 44



Glomerular Mechanisms

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Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0276. Identify the filtration barriers, if any, which impede the filtration of H₂O, Na⁺, inulin, albumin, and red blood cells.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0277. Define the filtration coefficient at the glomerular capillary, describe the membrane properties that contribute to it, and explain its role in determining GFR.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0278. Given the capillary and Bowman's capsule hydrostatic and oncotic pressures, calculate the net filtration force at the glomerular capillaries.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0352. Predict the changes in glomerular filtration caused by increases or decreases in any of those pressures.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0284. Define renal blood flow, renal plasma flow, glomerular filtration rate, and filtration fraction and list typical values.

Learning Objectives

SOM.MK.II.BPM1.3.CPR.2.PHYS.0279. Describe the relative resistances of the afferent and efferent arterioles and the effects on renal blood flow and GFR of selective changes in each.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0280. Describe the myogenic and tubuloglomerular feedback mechanisms that mediate the autoregulation of renal plasma flow and glomerular filtration rate.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0281. Predict the changes in net filtration force that occur as blood travels along the glomerular capillary and colloid osmotic pressure increases.

SOM.MK.II.BPM1.3.CPR.2.PHYS.0282. Explain the effects of changes in peritubular capillary hydrostatic and colloid osmotic pressures on net proximal tubular fluid reabsorption.

Recommended Readings

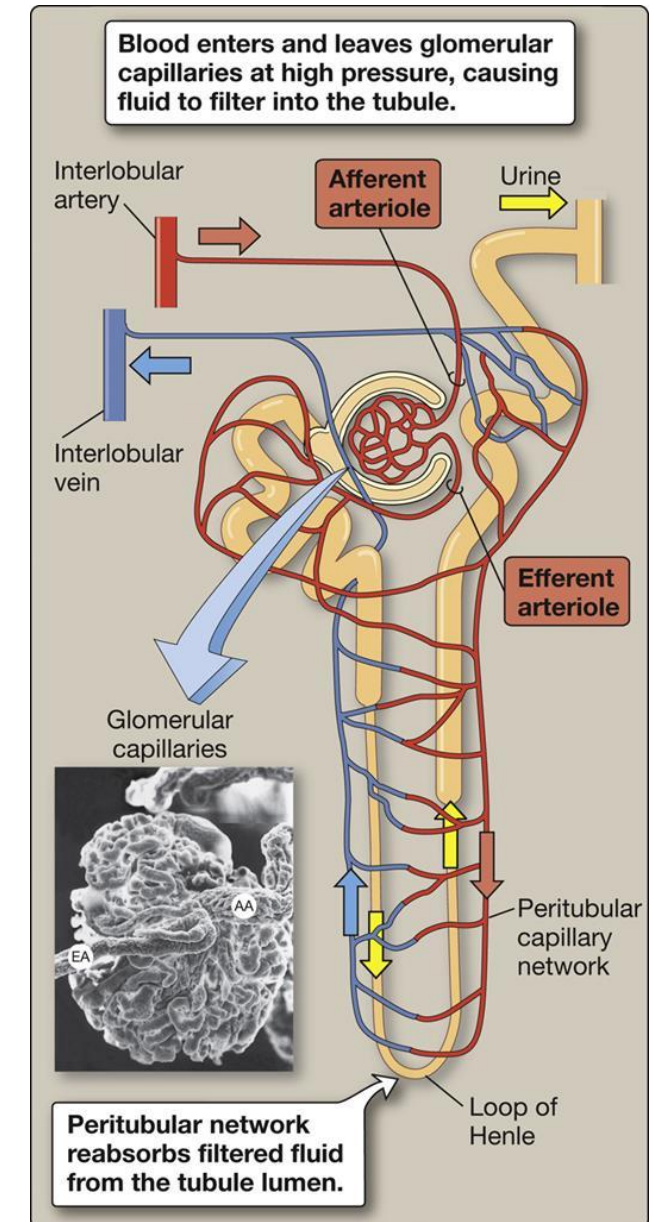
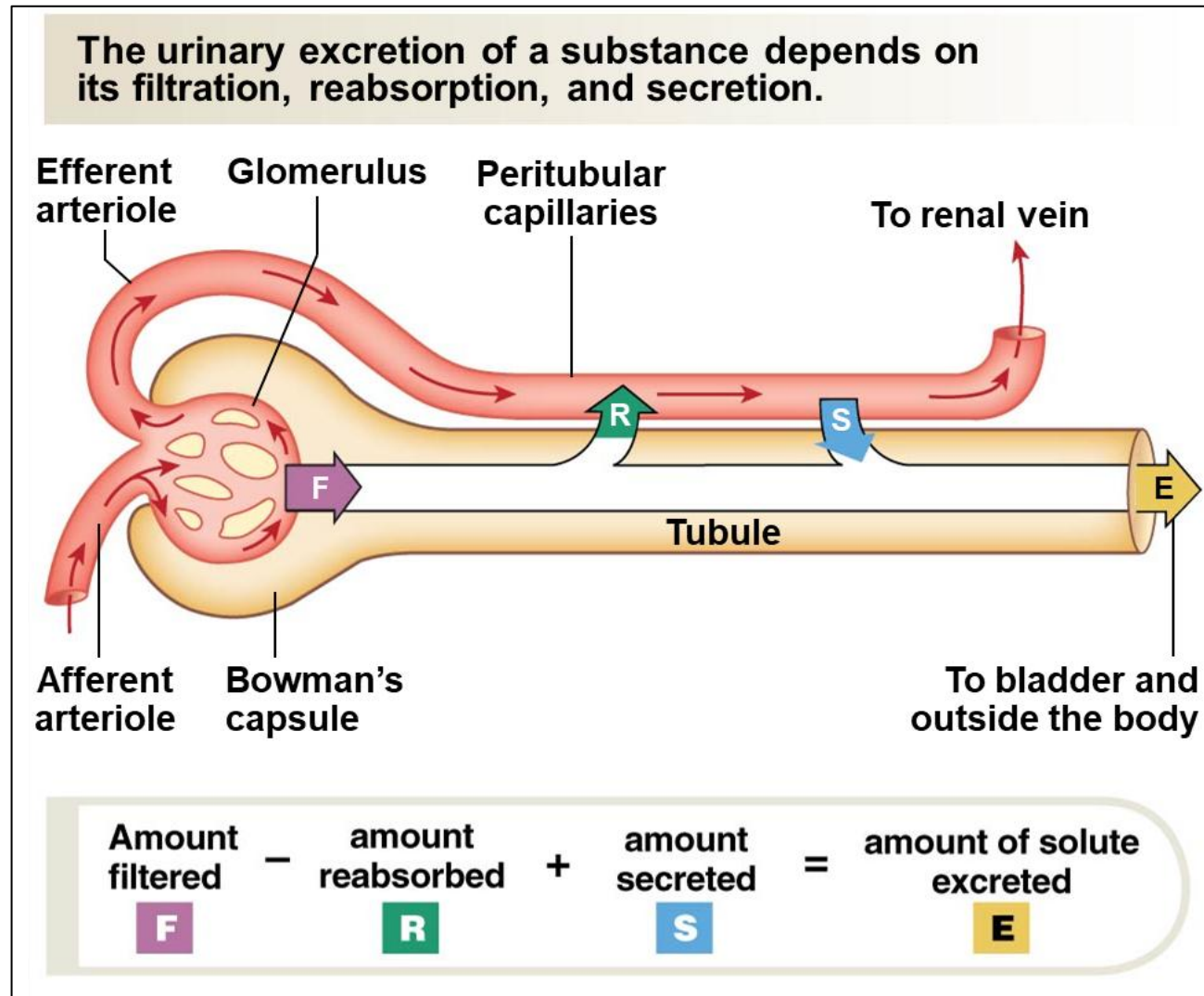
Chapter 22: Kidney Functions. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Chapter 23: Regulation of Fluid and Electrolyte Balance. Medical Physiology: Principles for Clinical Medicine, 6e. Rodney A. Rhoades, David R. Bell.

Chapter 27: Glomerular Filtration, Renal Blood Flow, and Their Control. Guyton and Hall Textbook of Medical Physiology, 14th Edition. John E. Hall PhD and Michael E. Hall MD, MS.

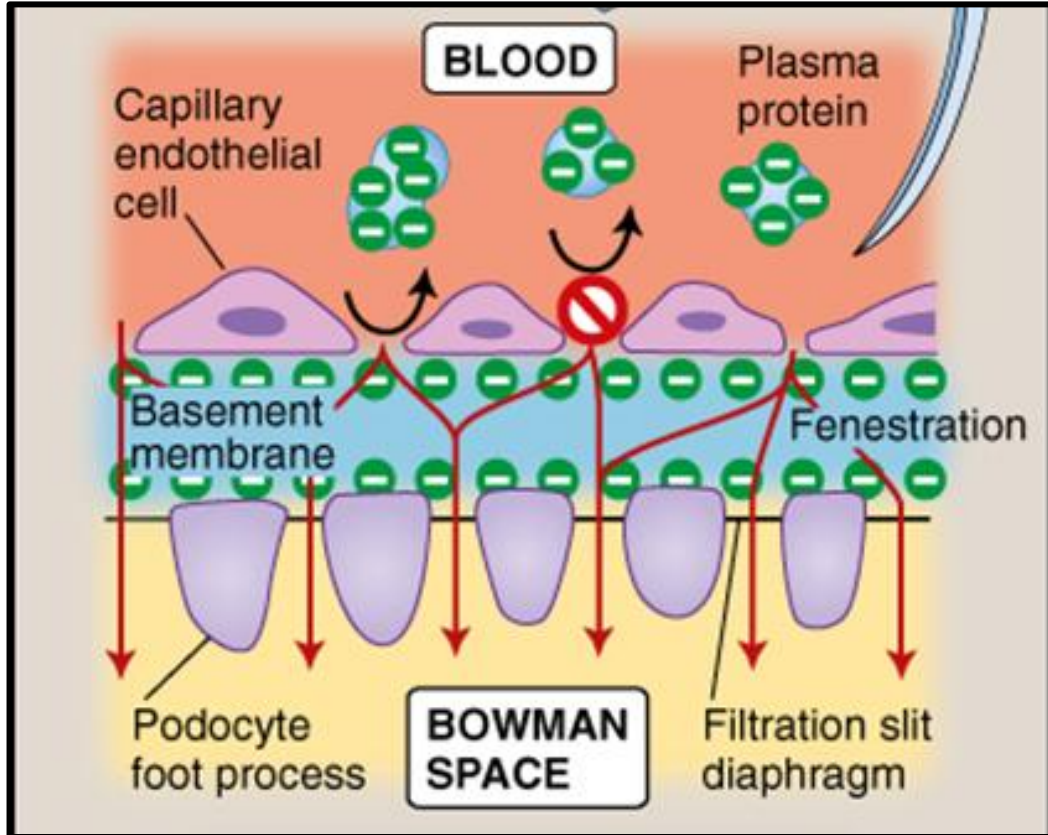
Slides with blue background are the supplemental information of the previous slides and thus, might not be explained again during the lecture time. However, please read the slides for clarification as they are equally important to know

Glomerular Filtration: First Step of Urine formation



Properties of Glomerular Filtration Barrier

- Filtration Barrier is made by 3 structures: Fenestrated Endothelium, Basement Membrane (BM), and Foot processes of the podocytes.
- Negatively charged molecules are filtered less easily than positively charged molecules of equal molecular size because of negative charges of the BM.



Filterability of Substances by
Glomerular Capillaries Based on Molecular Weight

Substance	Molecular Weight	Filterability
Water	18	1.0
Sodium	23	1.0
Glucose	180	1.0
Inulin	5500	1.0
Myoglobin	17,000	0.75
Albumin	69,000	0.005

Properties of Glomerular Filtrate

- Glomerular filtrate is an Ultrafiltrate.
- Ultrafiltrate contains all constituents of plasma EXCEPT plasma proteins.
- Substances that have the same concentration in both plasma and ultrafiltrate are called **Freely Filtered substances**. Examples: Glucose, Sodium, Potassium, Bicarbonate.

Elements of Glomerular Filtration Barrier

Ultrafiltration of blood across the glomerular capillaries into Bowman's space is the first step in urine formation.

Three major components of the Filtration Barrier are:

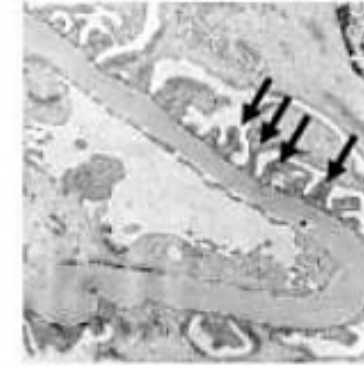
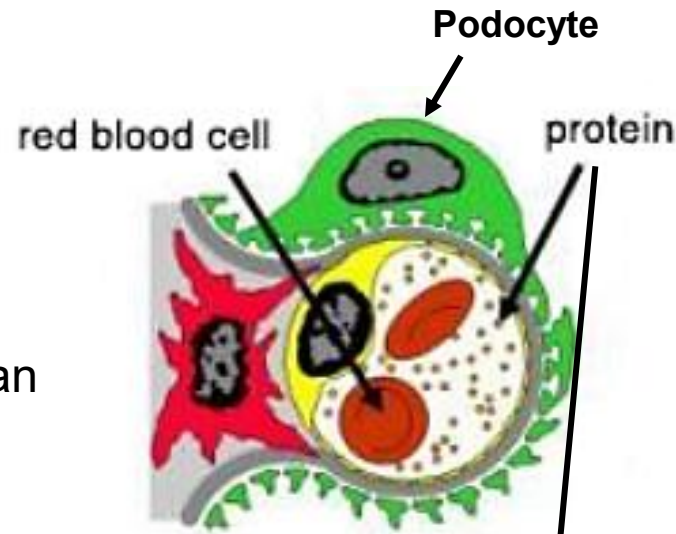
- 1. Endothelial fenestration of the glomerular capillaries**
 - Numerous >70-90 (nm or Å)
- 2. Glomerular basement membrane**
 - Fused basal lamina of the endothelial cells and podocytes
 - Heparin sulphate repels negatively charged molecules.
- 3. Podocyte layer** (visceral layer of the Bowman's capsule)
 - Modified epithelial cells called podocytes, highly ramified. Their foot processes make filtration slits (~40 nm gaps).

Properties of Glomerular Filtration Barrier

- Filterability of solutes is inversely related to their size.
- A filterability of 1 means the substance is filtered as **freely as water**, whereas filterability of 0.75 means the substance is filtered only 75% as rapidly as water.
- As molecular weight of the molecule approaches that of albumin, the filterability approaches zero.
- Albumin is about 6 nm whereas the pores of the glomerular membrane are about 8 nm. Even then, albumin is **NOT FILTERED** because of its negative charges.
- Negatively charged large molecules are filtered less easily than positively charged molecules of equal molecular size.

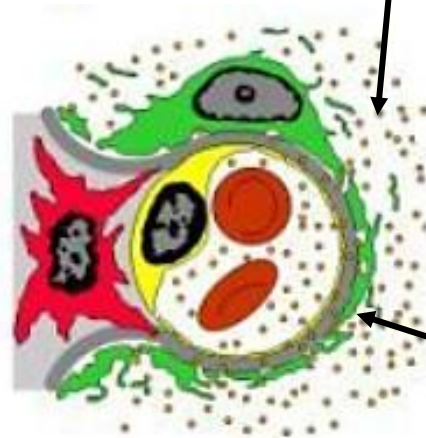
Filtration Barrier: Changes in Glomerular Diseases

Normal: Glomerular filtration barrier does not allow plasma proteins to get into the Bowman space and absent in urine.



By electron microscopy, a normal glomerular capillary has separate foot processes (arrows).

In **glomerular disease**, plasma proteins leak into the Bowman space and appear in urine because of injured filtration barrier. Urine study shows **Proteinuria** (protein in urine).



In a glomerular disease called minimal change disease, filtration barrier has fused foot processes (arrow).

Glomerular Filtration Rate (GFR)

Def: GFR is the volume of plasma filtered (ultrafiltrate) in unit time.

Expressed in ...mL per minute or Liter/day.

Formula to calculate GFR:

$$\text{GFR} = K_f \times \Delta P$$

K_f = Filtration coefficient (Normal: 12.5mL/min/mmHg)

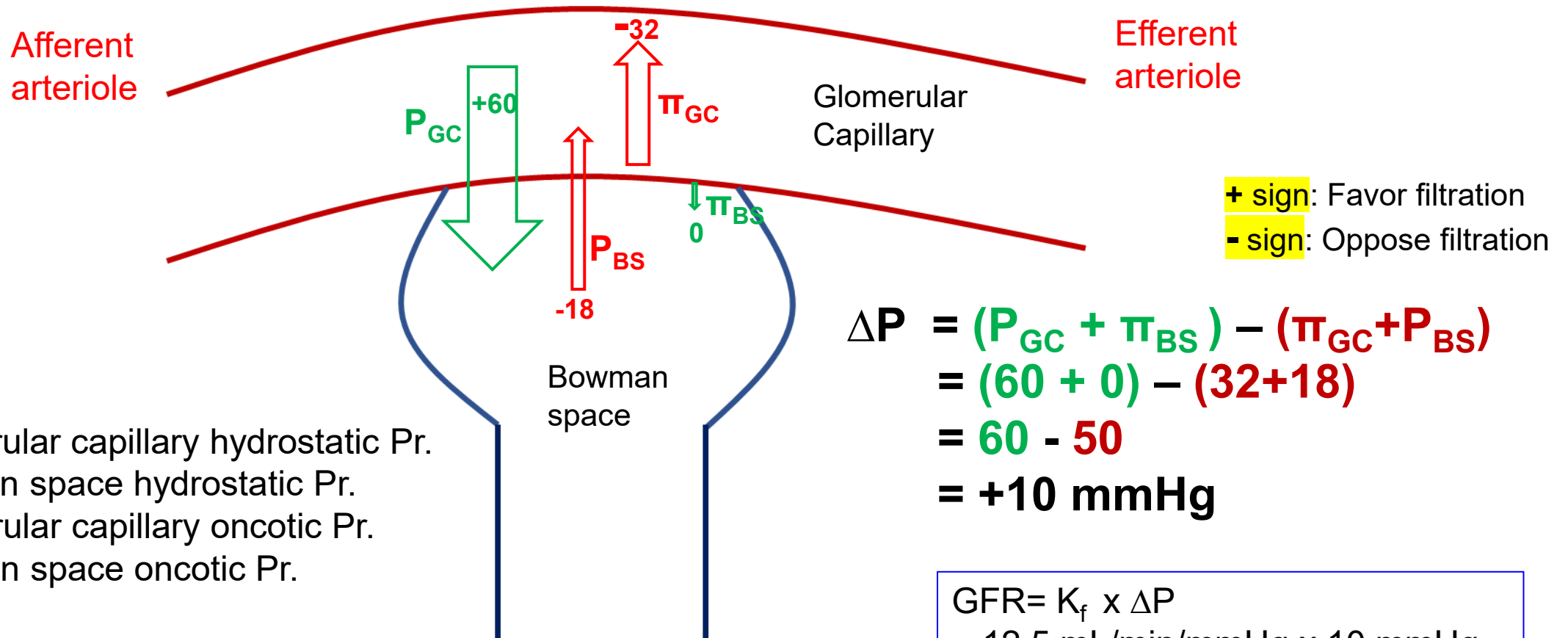
ΔP = Net Filtration Pressure (NFP)

Normal GFR value: 125 mL/min.

Glomerular Filtration Rate (GFR)

ΔP (Net Filtration Pressure, NFP) calculation

ΔP = Pressures that favor filtration - Pressures that oppose filtration



P_{GC} = Glomerular capillary hydrostatic Pr.
 P_{BS} = Bowman space hydrostatic Pr.
 π_{GC} = Glomerular capillary oncotic Pr.
 π_{BS} = Bowman space oncotic Pr.

$$\begin{aligned}\Delta P &= (P_{GC} + \pi_{BS}) - (\pi_{GC} + P_{BS}) \\ &= (60 + 0) - (32 + 18) \\ &= 60 - 50 \\ &= +10 \text{ mmHg}\end{aligned}$$

$$\begin{aligned}\text{GFR} &= K_f \times \Delta P \\ &= 12.5 \text{ mL/min/mmHg} \times 10 \text{ mmHg} \\ &= 125 \text{ mL/min}\end{aligned}$$

Glomerular Filtration Rate (GFR)

- **$GFR = K_f \times \Delta P$**

- GFR is the net fluid flow (protein-free) across the glomerular capillary wall . Normal GFR value for both kidneys is about 125 ml/min.

GFR is influenced by two factors:

- **Filtration coefficient (K_f)**
 - **Surface area** of the glomerular capillaries available for filtration.
 - **Permeability** of the filtration barrier.
- **Net filtration pressure (ΔP)**
 - Reduced GFR indicates renal insufficiency. A progressive decline in GFR occurs in chronic kidney disease.

Starling's forces (pressures):

- Hydrostatic pressure (P) → Push water
- Oncotic pressure (π) → Pull water.
- π is influenced by the presence of protein, specifically albumin

Forces favoring filtration

- Hydrostatic Pr in glomerular capillary ($P_{GC} = 60$ mmHg).
- Oncotic Pr in Bowman's space ($\pi_{BS} = 0$ mmHg).

Forces opposing filtration

1. Hydrostatic Pr Bowman's space ($P_{BS} = 18$ mmHg).
2. Oncotic Pr in glomerular capillary ($\pi_{GC} = 32$ mmHg).

What drives GFR is P_{GC} = Hydrostatic pressure in glomerular capillaries

FILTRATION FRACTION (FF)

- Filtration fraction (FF) is the **volume of protein-free fluid** that is filtered into the Bowman's space, out of the renal plasma flow (RPF) in the kidney.
- **$FF = GFR/RPF$**

$GFR = 125\text{mL/min}$ and $RPF = 625\text{mL/min}$

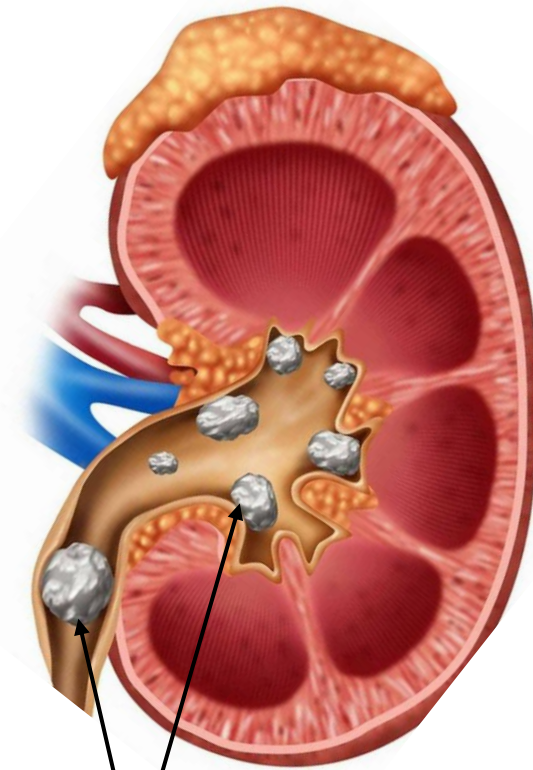
$FF = 125/625 = 20\% (0.2)$

- 1/5 (20%) of the renal plasma in the glomerular capillaries filters into the Bowman's capsule as **ultrafiltrate**. This loss of protein-free fluid concentrates plasma proteins in the blood leaving the glomerular capillaries and entering the efferent arterioles.

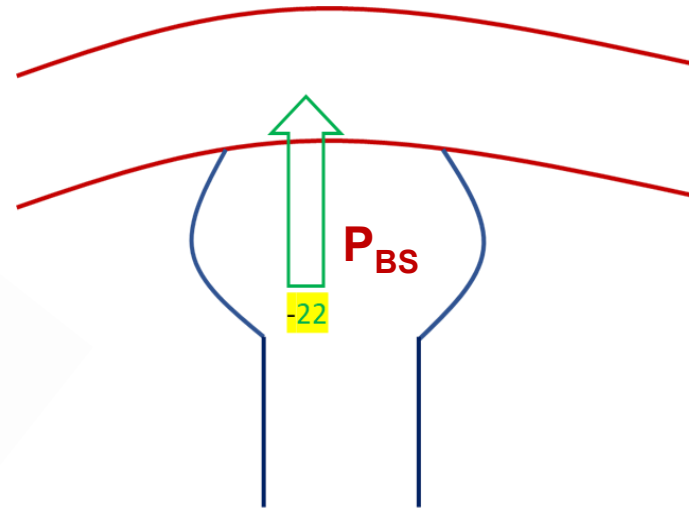
Salient features of Filtration Coefficient (K_f)

- K_f is a measure of the product of **permeability** and **surface area** of the glomerular filtration barrier.
- Increased K_f raises GFR and decreased K_f reduces GFR. However changes in K_f probably do not provide a primary mechanism for the normal day to day regulation of GFR.
- Diseases (Kidney diseases, uncontrolled hypertension and diabetes mellitus) **lower K_f** by reducing the number of functional glomerular capillaries (thereby reducing the surface area for filtration) or by increasing the thickness of the filtration barrier (reducing permeability).
- Mesangial cells in the glomerulus influence K_f . Contraction of these cells reduce capillary surface area, reduce K_f and thus reduce GFR. The role of mesangial cells is minor compared with that of glomerular arterioles.

Clinical Correlate: Kidney stones (Nephrolithiasis) may reduce GFR



Stones



Keep values of all other pressures same as normal and **increase P_{BS}** and apply in the GFR formula

$$\begin{aligned}\Delta P \text{ (mmHg)} &= (P_{GC} + \pi_{BS}) - (\pi_{GC} + P_{BS}) \\ &= (60 + 0) - (32 + 22) \\ &= +6 \text{ mmHg}\end{aligned}$$

Reduced ΔP means a reduction in GFR

$$\text{GFR} = K_f \times \Delta P$$

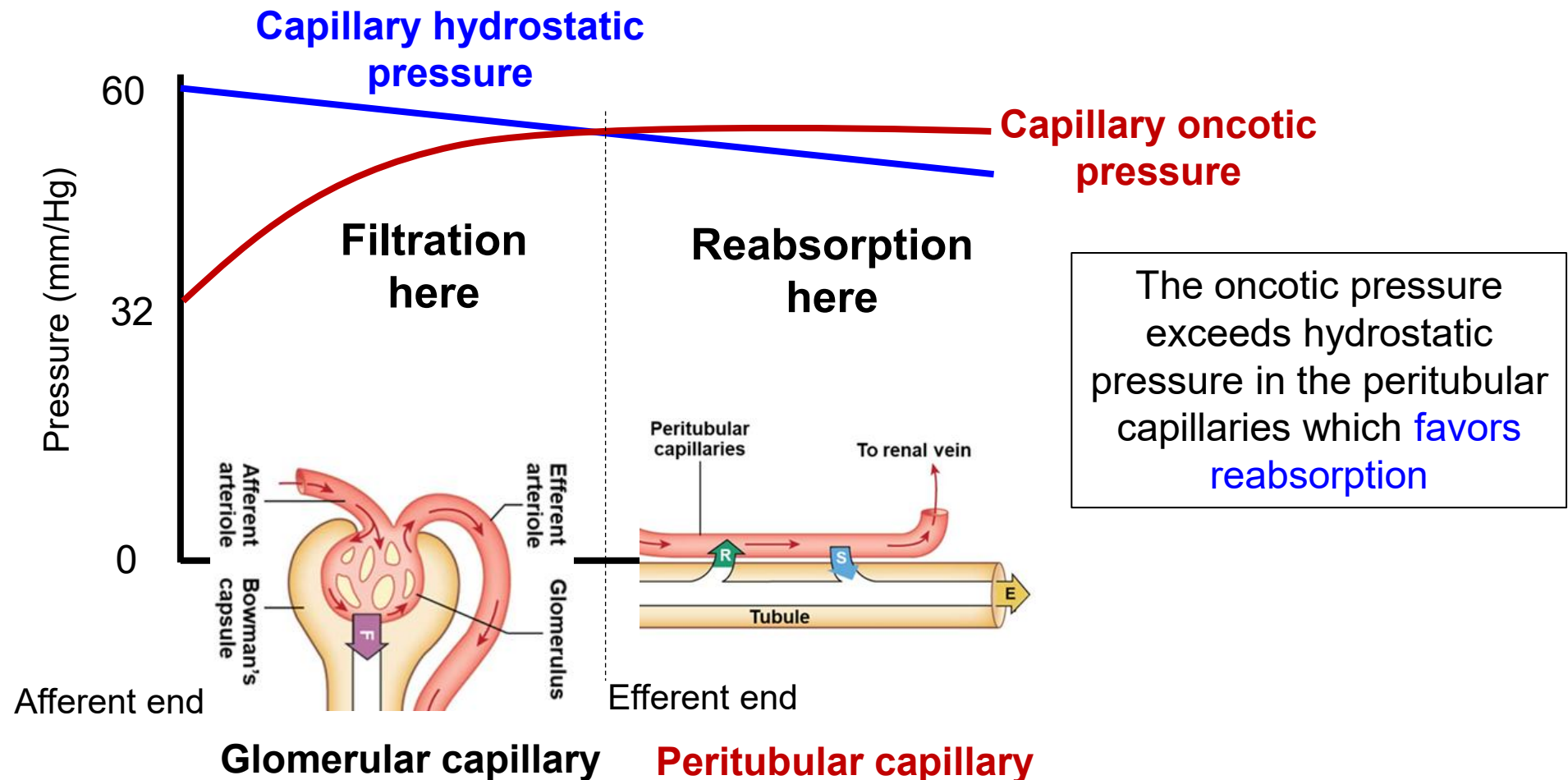
Clinical Correlate: Examples of Kidney Diseases affecting the glomerular functions

- Nephrotic Syndrome is a glomerular disease in which a massive amount of plasma protein is filtered and excreted in urine. The amount of albumin lost is more than 3 g/day which will greatly reduce the plasma oncotic pressure and produce generalized edema.
- Nephritic Syndrome is a glomerular disease in which red cells along with plasma proteins escape through the glomerulus and appear in urine. Presence of red cells in urine is called **hematuria**. These patient will have edema and anemia.
- Nephrolithiasis (Kidney stones) obstruct the flow of urine and rise hydrostatic pressure in the Bowman space.

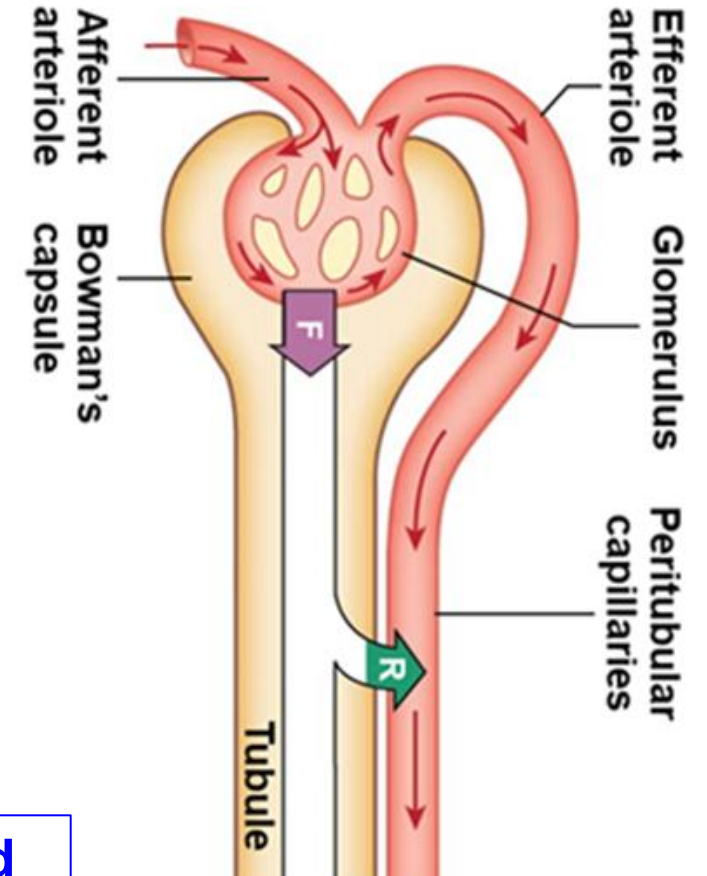
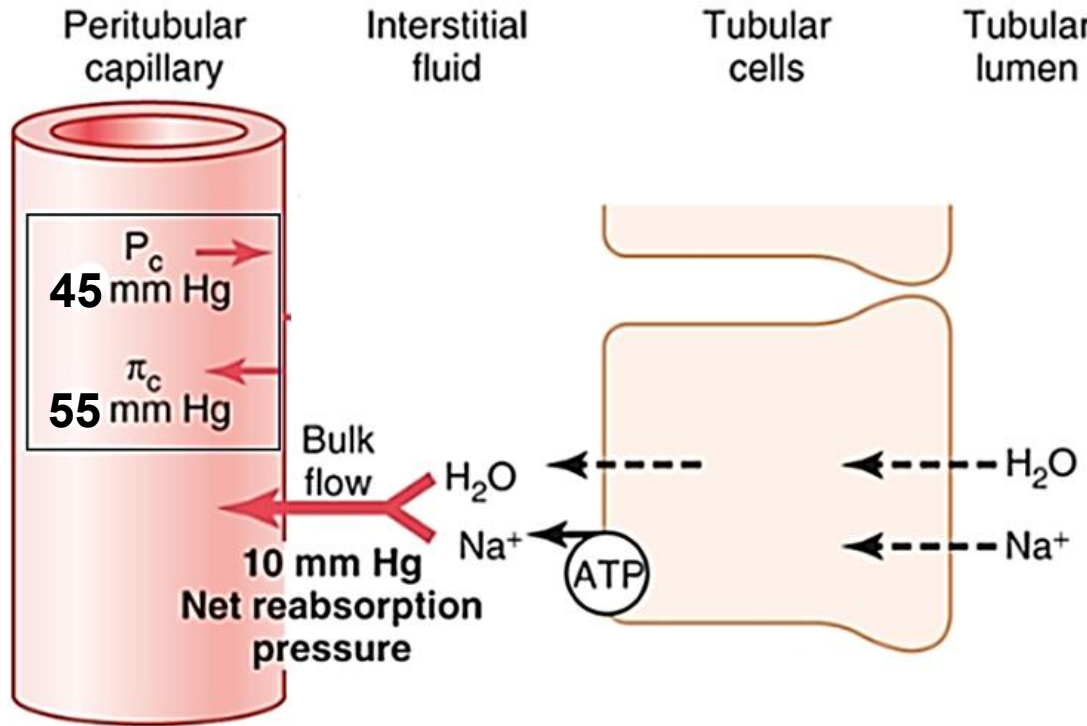
Reduced glomerular filtration results in reduction in GFR in all these conditions.

Glomerular capillary oncotic pressure & Its Importance

Glomerular capillary **oncotic pressure increases** significantly as blood passes from the afferent arteriole and exits through the efferent arteriole.



Rise in capillary oncotic pressure favors reabsorption in the peritubular capillaries



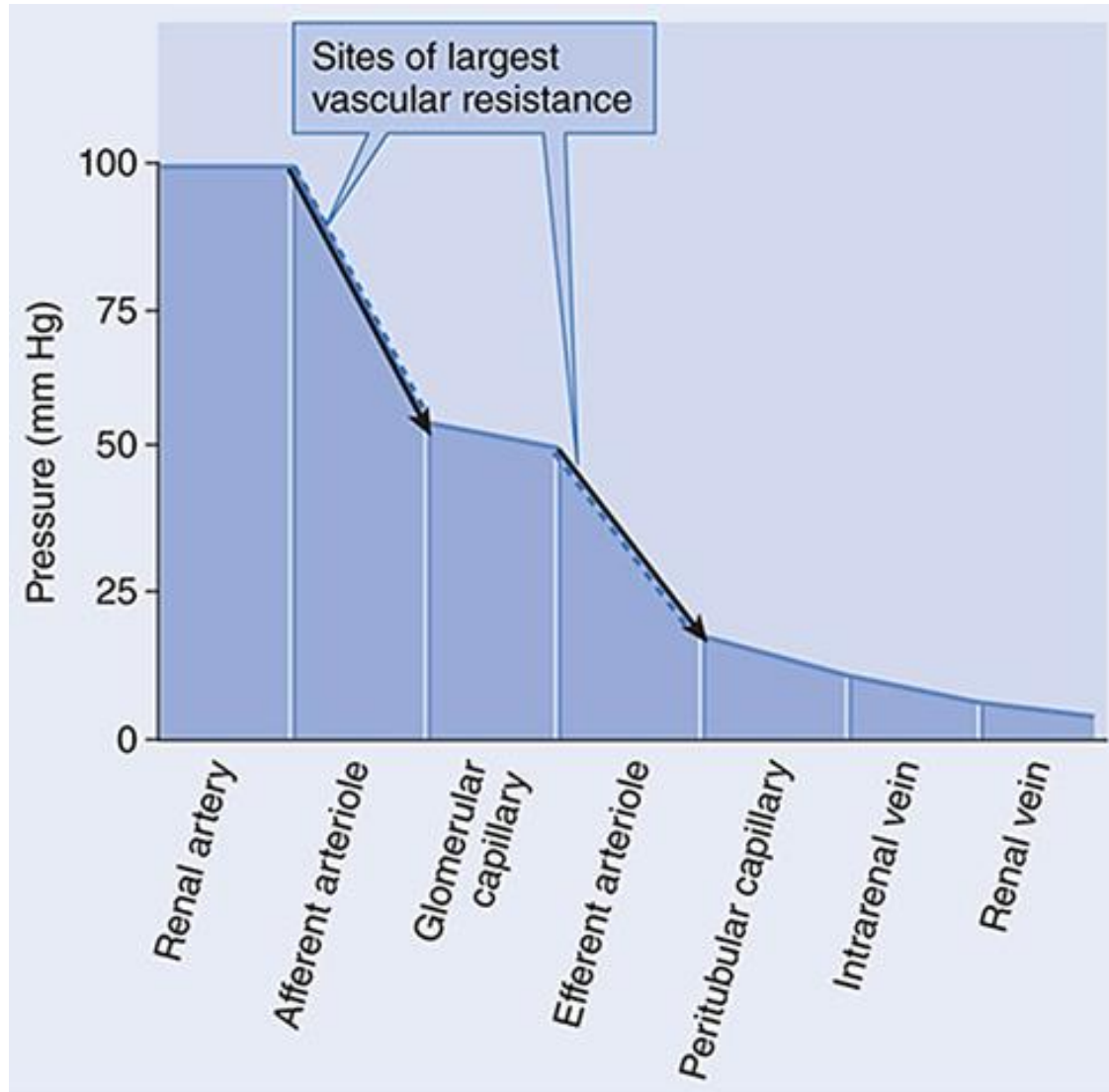
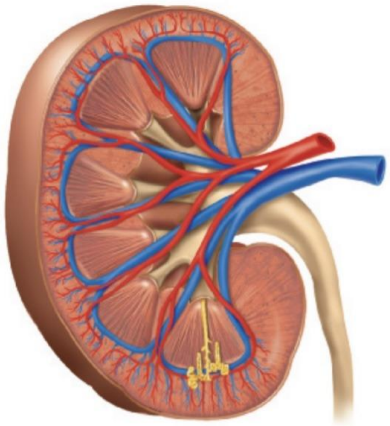
Net reabsorption pressure allows movement of reabsorbed solutes back into peritubular capillaries in the kidney

P_c = Hydrostatic pressure of peritubular capillary
 π_c = Oncotic pressure of peritubular capillary

Changes in Starling's Forces as blood moves in the glomerular and peritubular capillaries and their importance

- Starling's forces change as blood passes along the glomerular capillary from the afferent arteriole to the efferent arteriole and to the peritubular capillaries.
- Glomerular capillary oncotic pressure increases as blood passes from the afferent arteriole through the glomerular capillaries to the efferent arteriole as it loses protein-free fluid into the Bowman space.
- Oncotic pressure exceeds hydrostatic pressure in the peritubular capillaries.
- Increased oncotic pressure in peritubular capillary favors reabsorption of interstitial fluid back into the blood by bulk flow.
- This bulk flow carries back the fluid and solutes that are reabsorbed in the renal tubules.
- Two sets of capillaries perform two different functions in the kidney because of these changes in the Starling's forces: The glomerular capillaries are responsible for filtration of plasma and the peritubular capillaries are responsible for reabsorption of the fluid and the solutes back into the plasma.

Renal Hemodynamics: General Rules



Blood pressure decreases as blood flows through the renal vascular network

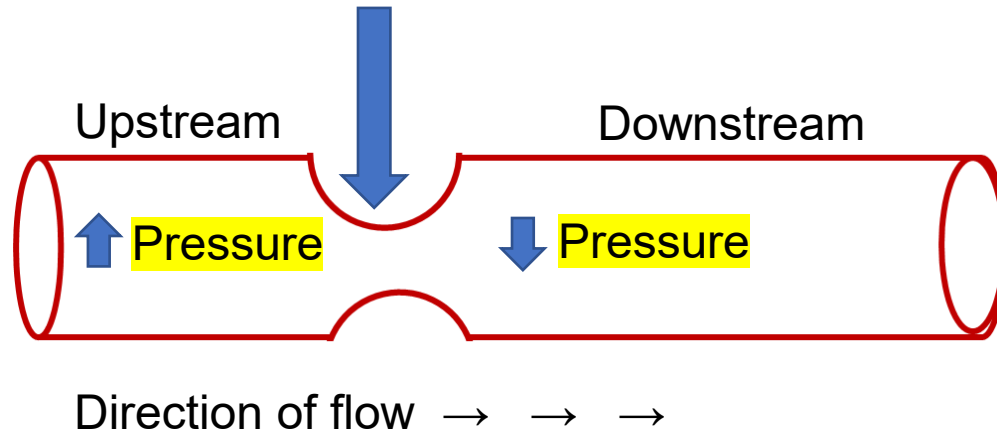
Greatest pressure drops occur at afferent and efferent arterioles

Vasoconstriction of either afferent or efferent arterioles increases **renal vascular resistance** and thus reduces renal blood flow

Vasodilation of either afferent or efferent arterioles decreases resistance and therefore increases renal blood flow

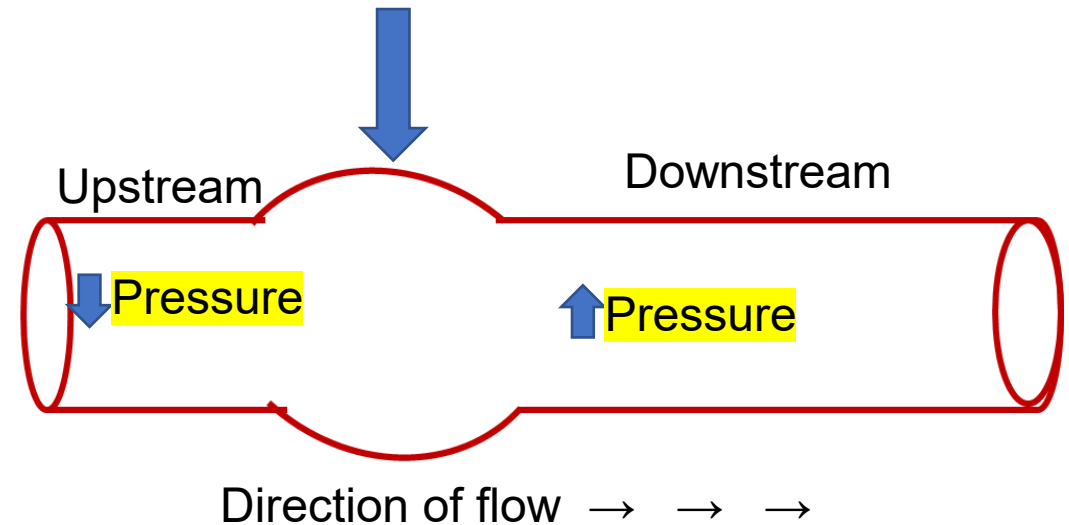
Renal Hemodynamics General Rules

Vasoconstriction site



1. Total flow decreases
2. Pressure upstream increases
3. Pressure downstream decreases

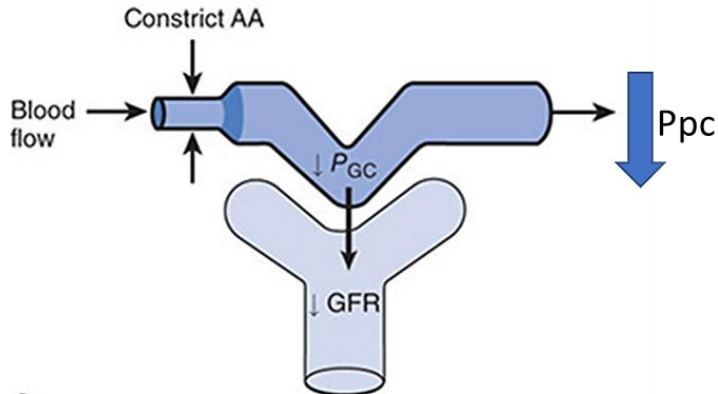
Vasodilation site



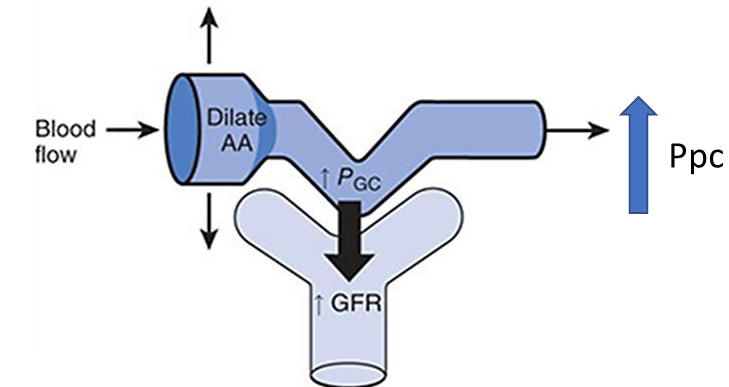
1. Total flow increases
2. Pressure upstream decreases
3. Pressure downstream increases

Control of renal blood flow and GFR by altering the resistance of afferent and efferent arteriole

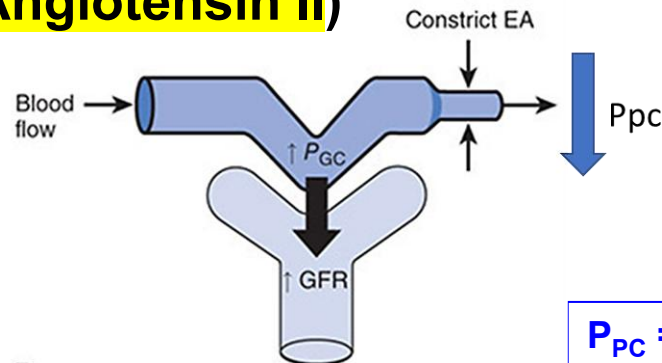
Afferent Arteriole- Vasoconstriction



Afferent Arteriole- Vasodilation (examples: Prostaglandins, Bradykinin)

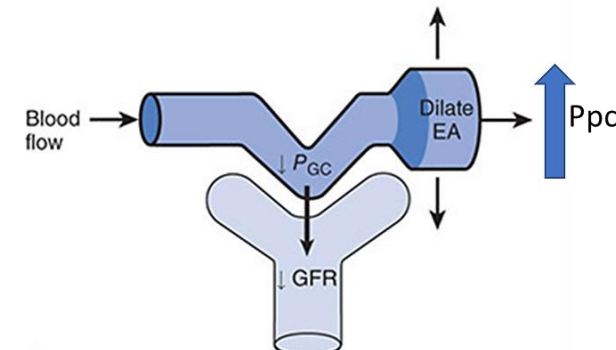


Efferent Arteriole- Vasoconstriction (Example: Angiotensin II)



P_{PC} = Pressure in peritubular capillary
 P_{GC} = Pressure in glomerular capillary

Efferent Arteriole- Vasodilation



Afferent Arteriole- Vasoconstriction

1. Decrease in hydrostatic pressure of glomerular capillaries
2. Decrease in GFR
3. Decrease in renal plasma/blood flow

Afferent Arteriole- Vasodilation(Prostaglandins, PGE₂ and PGI₂, Bradykinin)

1. Increase in hydrostatic pressure of glomerular capillaries
2. Increase in GFR
3. Increase in renal plasma/blood flow

Efferent Arteriole- Vasoconstriction (Angiotensin II)

1. Increase in hydrostatic pressure of glomerular capillaries
2. Increase in GFR
3. Decrease in renal plasma/blood flow

Efferent Arteriole- Vasodilation

1. Decrease in hydrostatic pressure of glomerular capillaries
2. Decrease in GFR
3. Increase in renal plasma/blood flow

Summary of isolated constrictions or dilations of the afferent and efferent arterioles

	GLOMERULAR CAPLLARY PRESSURE	RENAL PLASMA & BLOOD FLOW	FILTRATION FRACTION (FF) FF= GFR/RPF
CONSTRICT EFFERENT	↑	↓	↑
DILATE EFFERENT	↓	↑	↓
CONSTRICT AFFERENT	↓	↓	No Change
DILATE AFFERENT	↑	↑	No Change

Examples of clinical conditions that can decrease GFR, and the mechanism involved

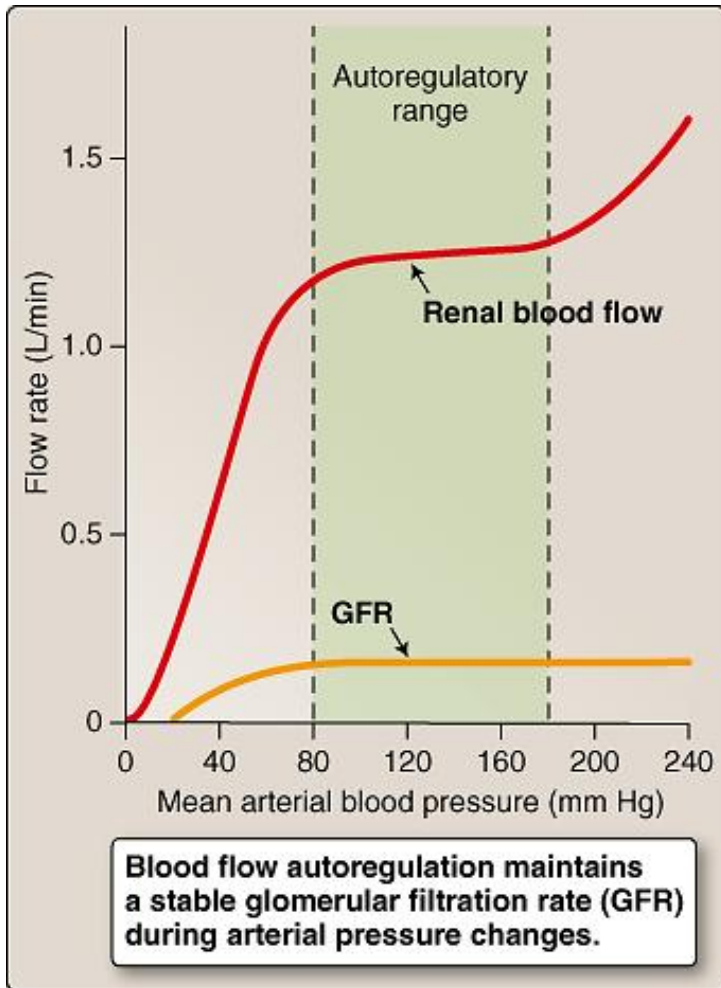
Physical Determinants*	Physiological/Pathophysiological Causes
$\downarrow K_f \rightarrow \downarrow \text{GFR}$	Renal disease, diabetes mellitus, hypertension
$\uparrow P_B \rightarrow \downarrow \text{GFR}$	Urinary tract obstruction (e.g., kidney stones)
$\uparrow \pi_G \rightarrow \downarrow \text{GFR}$	$\downarrow$ Renal blood flow, increased plasma proteins
$\downarrow P_G \rightarrow \downarrow \text{GFR}$ $\downarrow A_P \rightarrow \downarrow P_G$	$\downarrow$ Arterial pressure (has only a small effect because of autoregulation)
$\downarrow R_E \rightarrow \downarrow P_G \rightarrow \downarrow \text{GFR}$	$\downarrow$ Angiotensin II (drugs that block angiotensin II formation)
$\uparrow R_A \rightarrow \downarrow P_G \rightarrow \downarrow \text{GFR}$	$\uparrow$ Sympathetic activity, vasoconstrictor hormones (e.g., norepinephrine, endothelin)

A_P , systemic arterial pressure; GFR, glomerular filtration rate; K_f , glomerular filtration coefficient; P_B , Bowman's capsule hydrostatic pressure; π_G , glomerular capillary colloid osmotic pressure; P_G , glomerular capillary hydrostatic pressure; R_A , afferent arteriolar resistance; R_E , efferent arteriolar resistance.

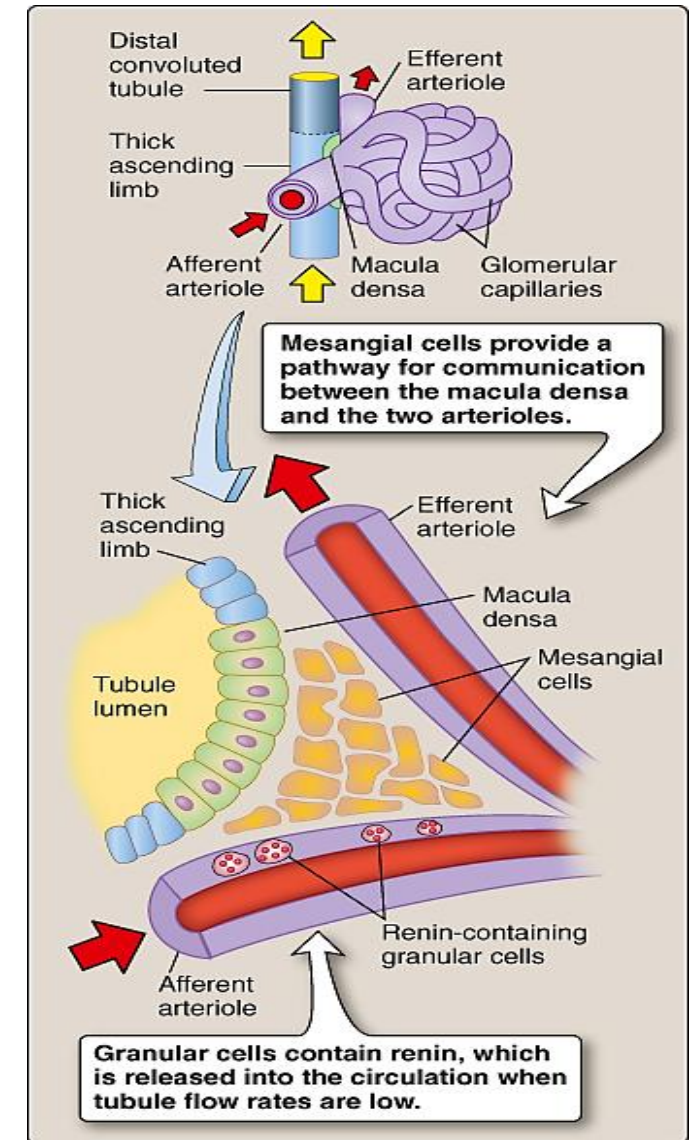
Take home exercise:

Explain with reasoning for the decreased GFR in each of the listed physiological and pathological conditions

Kidney- Blood flow and Autoregulation



Macula densa cell output and arteriolar response depends on whether tubular fluid flow rates are high or low



Autoregulation in the kidneys maintains a **relatively constant GFR** and allows precise control of renal excretion of water and solutes.

Kidney: Blood flow and Autoregulation

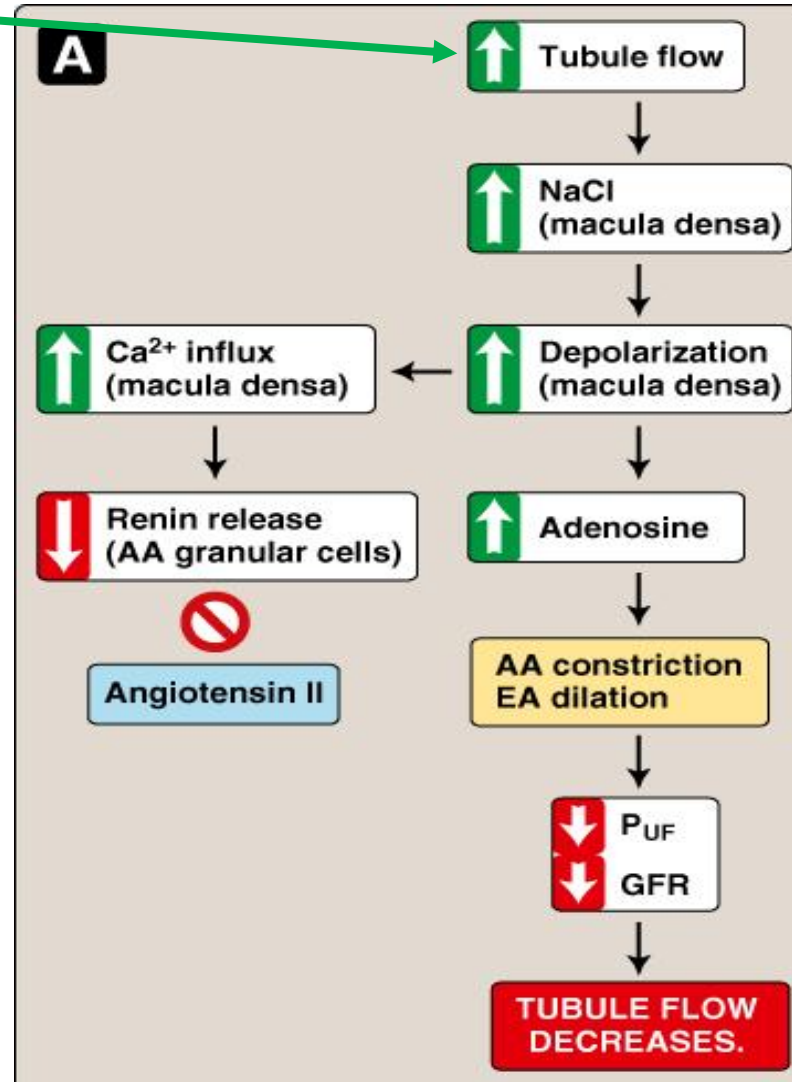
- Autoregulation in the kidneys maintains a **relatively constant GFR** and allows precise control of renal excretion of water and solutes.
- If GFR too high → Needed substances cannot be reabsorbed quickly enough and is lost in the urine.
- If GFR too low → Too much reabsorption occurs inclusive of waste products such as Urea.
- RBF remains relatively stable over a mean arterial pressure (MAP) range of ~80 to 180 mm Hg.
- Autoregulation is primarily accomplished due to changes in the resistance of the afferent arteriole by two mechanisms:
 1. Myogenic mechanism: The intrinsic property of smooth muscle is to contract when stretched. High blood pressure stretches the smooth muscles of the afferent arterioles which cause vasoconstriction and reduce the blood flow back to normal.
 2. Tubuloglomerular feedback (TGF): TGF is an autoregulatory mechanism mediated by the juxtaglomerular apparatus (JGA) that adjusts RBF and GFR to optimize fluid flow through the renal tubule.

Tubuloglomerular Feedback: Autoregulation of GFR

↑ **GFR**

Signals from the renal tubule in each nephron feed back to affect filtration in its glomerulus.

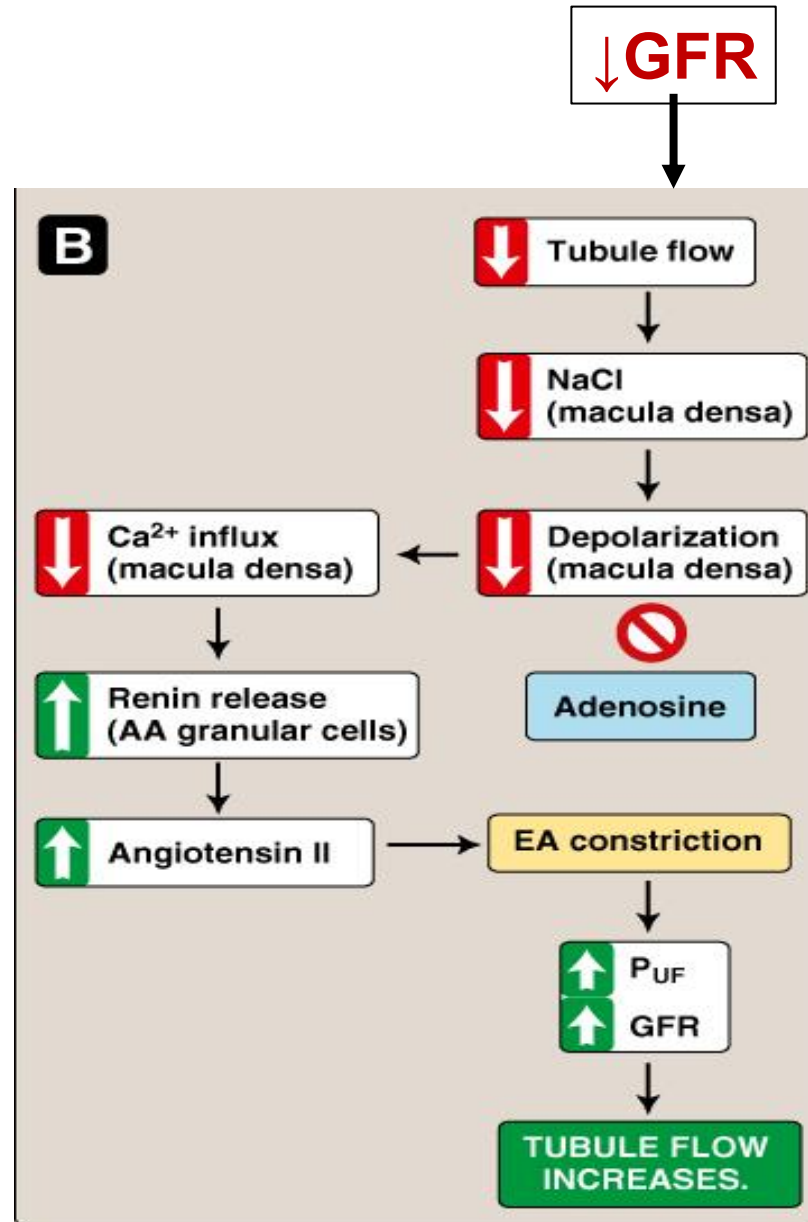
Increase in tubular flow reduces the GFR in the same nephron and, conversely, a decrease in tubular flow increases the GFR



Conditions: Transient increase in BP or blood volume

AA = afferent arteriole; EA = efferent arteriole; GFR = glomerular filtration rate; P_{UF} = capillary hydrostatic ultrafiltration pressure.

Tubuloglomerular Feedback: Autoregulation of GFR



A decrease in tubular flow increases the GFR

Condition: Transient fall in BP or blood volume

AA = afferent arteriole; EA = efferent arteriole; GFR = glomerular filtration rate; P_{UF} = capillary hydrostatic ultrafiltration pressure.

Summary

- Glomerular filtration barrier has **three layers**, and it restricts filtration of large molecules such as albumin. **Negative surface charge** on the filtration barrier basal lamina restricts filtration of negatively charged solutes more than positively charged solutes.
- GFR is determined by the **filtration coefficient** (K_f) and **net filtration pressure** (NFP). NFP varies mainly with hydrostatic and oncotic pressures in the glomerular capillaries.
- Independent constriction and dilation of the afferent and efferent arterioles permits independent control of glomerular filtration, renal blood flow, and filtration fraction.
- **Autoregulation** of blood flow keeps GFR within limits in the face of large variations in arterial pressure.

Do It by Yourself : Read the case history and answer the questions

A 32-year-old man is brought to the emergency department because of a 2-week history of puffy face, reduced urine output, and lethargy. He was Covid-19 positive three weeks ago and currently declared negative. His pulse is 84/min and blood pressure 144/86 mm Hg. Physical examination shows pitting edema in the legs. Urinalysis shows massive proteinuria (3.8 g/day) but no red blood cells. Serum studies show albumin concentration of 2.3 g/dL (normal: 3.4–4.8 g/dL) and hyperlipidemia. A renal biopsy is performed. Light microscopic examination does not show any changes. However, electron microscopic examination shows fused foot processes in the glomerulus. He is diagnosed with minimal change disease.

1. What is massive proteinuria?
2. Is it a nephrotic or nephritic syndrome and why?
3. What is the explanation for the proteinuria?
4. How does the body responses to this abnormally high urinary protein loss?
5. What is the reason for the edema?

Do It by Yourself: Answers

1. What is massive proteinuria?

Ans: The urinary excretion of more than 3.5 g/day of proteins (mainly albumin).

2. Is it a nephrotic or nephritic syndrome and why?

Ans: It is nephrotic syndrome as the proteinuria is massive and no red cells in urine microscopy. Three classical features of nephrotic syndrome are: massive proteinuria, hypoalbuminemia (<3 g/dL) with generalized edema, and hyperlipidemia.

3. What is the explanation for the proteinuria?

Ans: Proteinuria is a consequence of an abnormally high permeability of the glomerular filtration barrier to the plasma proteins. There is an increased physical size of “holes” or pores in the filtration slit diaphragms and a loss of fixed negative charges from the glomerular basement membrane. All these factors contribute to a heavy loss of proteins.

4. How does the body responses to this abnormally high urinary protein loss?

Ans: Massive proteinuria lowers capillary oncotic pressure. This stimulates the liver to synthesize proteins at a higher rate. In addition to albumin, liver synthesizes higher amounts of lipoproteins that results in hypercholesterolemia and hyperlipidemia in the patient.

5. What is the reason for the edema?

The edema is due to altered Starling forces in the capillaries in the tissues and renal retention of salt and water. The decrease in plasma albumin (or protein) decreases the plasma oncotic pressure. Reduced oncotic pressure favors fluid movement out of the systemic capillaries (it means, capillaries throughout the body) into the interstitial compartment. The edema occurs throughout the body, but noticeable in the soft tissues like skin around the eyes (periorbital edema).

Any Questions ?

How to reach me: Either walk into my office or email.

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Other References

1. Costanzo Physiology: Linda S. Costanzo PhD, Seventh Edition, Chapter 6, Renal Physiology.
2. Netter's Essential Physiology, Third Edition: Mulroney, Susan E., PhD, Chapter 17, Overview, Glomerular Filtration, and Renal Clearance.
3. Lippincott Illustrated Reviews: Physiology, Third Edition: Robin R. Preston, Thad E. Wilson. Chapter 25, Filtration and Micturition.